

Iman Kahssay

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EDUCATION

M.S. in Computer Science, University of California, Davis **3.8/4.0 GPA**, Jun 2026

Relevant Coursework: Software Engineering, AI/ML, Algorithm Design & Analysis, Data Visualization, Compilers, UX/UI

B.S. in Electrical Engineering & Computer Science, University of California, Berkeley May 2022

Relevant Coursework: Data Structures & Algorithms, AI, Discrete Math, Graphics & Imaging, Computer Security, UX/UI

SKILLS

Languages: Python, Java, C/C++, Kotlin, Go, SQL, HTML, XML, JavaScript

Tools: Flask, React, Node.js, Express, MongoDB, Git, NumPy, PyTorch/TorchVision, Postman, Android Studio, Figma

PROFESSIONAL EXPERIENCE

Researcher, Interactive Organisms Lab at UC Davis Sep 2024 - Jun 2026

- Designed and evaluated a microfluidic device with colorimetric biosensors to detect biomarkers in bodily fluids using Fusion 360 & FormLabs; tested prototypes for sensor performance and reliability
- Developed an Android app in Kotlin & XML using Figma and Android Studio; tested and validated its sensor data integration for real-time biomarker detection with **80% accuracy**, personalized health summaries, and recommended actions
- Led the full IRB and user study** process for 21 participants, from protocol development and recruitment through execution & analysis; findings **improved device usability by 20%**, informed app refinements, and resulted in a **co-first-author publication**
- Improved lab reproducibility and project execution by documenting device fabrication and app integration workflows, managing undergraduate researchers, and coordinating equipment, inventory, purchases, and out-of-state shipments

Teaching Assistant, UC Davis Sep 2024 - Jun 2026

- Supported **150+ students across 6 courses** in Python, Processing, MakeCode, Arduino, and Circuit Playground; developed lecture slides, coding exercises, and quizzes; coordinated 3D printing, vinyl/laser cutting, soldering, embedded systems labs
- Resolved 95% of software and hardware issues** involving debugging, environment setup, program logic, and embedded systems workflows while training students and grading **1,000+ technical assignments, exams, and projects**

Technical Account Analyst, Zain Psychological & Behavioral Health Services Jan 2022 - Sep 2024

- Built scalable RESTful APIs using Flask, Python, and MongoDB to support internal accounting workflows, enabling reliable data retrieval, CRUD operations, and integration with the accounting user interface
- Executed end-to-end testing in Postman, **improving user flow by 23% & page speed by 19%**

PROJECTS

FitBot | React, Node.js, Express, MongoDB, Python Mar 2025 - Jun 2025

- Built a personalized fitness analytics web app in a 3-person team, using Python and JavaScript, to compute and visualize calories burned and goal progress; leveraged research to **improve clarity by 45% and user engagement by 52%**
- Engineered a scalable MERN architecture, enabling real-time updates, backend computation, and interactive visualizations
- Implemented multi-feature calorie-burn algorithms using 6+ physiological and activity variables across 10K+ activity records, **increasing personalization and accuracy** of health insights **by 35%**

BugSwarm Automated Bug Reporter | Python, REST APIs, CI, LLM Sep 2025 - Dec 2025

- Collaborated on a 4-person team to build an end-to-end automated bug reporting pipeline in Python, transforming raw CI failure artifacts into structured, human-readable bug reports and **reducing manual bug documentation** effort by **~70%**
- Integrated BugSwarm and Travis CI via REST APIs to ingest and process CI failure artifacts, implemented log sanitization and diff summarization, and integrated an LLM (Google Gemini) to generate root-cause analyses and fix suggestions; **generated 100/100 reports with 0 runtime errors**, demonstrating system reliability and scalability

Cloth Simulation with Realistic Rendering | C++, OpenGL, GLSL Jan 2022 - May 2022

- Developed an interactive cloth simulation in C++ using a point mass and spring-based system, implementing physical constraints, numerical integration, and collision handling to accurately model cloth movement and reduce cloth clipping by **90%**
- Wrote GLSL programs to **improve visual & texture quality by 85%**, and developed two new shaders through research on new/improved shading models for textile materials, **optimizing visual quality by 80%**

Grammy's Winner Prediction | Java, Android SDK, XML Jan 2022 - May 2022

- Led a team of four to create an Android app where users can select, share, and discuss Grammy winner predictions with the public; implemented core application logic and UI components using Java & XML
- Tested and debugged workflows using unit tests and manual validation, **improving user flow by 51% and page speed by 43%**